



POLITECNICO
MILANO 1863

**SCUOLA DI INGEGNERIA INDUSTRIALE
E DELL'INFORMAZIONE**

EXECUTIVE SUMMARY OF THE THESIS

LAUREA MAGISTRALE IN AERONAUTICAL ENGINEERING

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Advisor: PROF. FEDERICO GHIOLDI

Co-advisor:

Academic year: 2024-2025



Figure 1: Caption

1. Introduction

This report delves into the aerodynamic interaction between two airfoils arranged in a tandem configuration, with the primary goal of maximizing the lift coefficient (C_L) of the front airfoil. The analysis is carried out under a variety of conditions, including the presence and absence of ground effect, providing insights into scenarios relevant to real-world applications such as vehicles operating near ground surfaces. By studying the pressure coefficient (C_p) distributions and variations in (C_L), the report highlights the role of geometric factors, boundary conditions, and their interplay in influencing aerodynamic

performance.

Config	CPU	RAM	Clock time
	Intel core i3-6006U	12Go	2.0 GHz

Table 1: Configuration of my computer

2. Guidelines

The study begins with an individual analysis of each airfoil to evaluate their aerodynamic performance and validate the methodology against thin airfoil theory and XFLR5 simulations. This step establishes a foundation for more complex investigations.

The focus then shifts to the tandem configuration, examining interactions between the front and rear airfoils, particularly under ground effect. The analysis highlights how ground proximity alters lift and pressure distributions, revealing key aerodynamic phenomena.

Building on these results, the report explores optimal configurations to maximize the front airfoil's lift coefficient (C_L) by varying angles of attack and relative positions. This process is crucial for efficient aerodynamic design.

Finally, the study transitions to practical application by applying the findings to a real indoor

model aircraft, bridging the gap between theory and real-world scenarios.

3. Study of each airfoil

The objective of this first section is to individually analyze each airfoil by calculating the pressure coefficient C_p and the lift coefficient C_L . I will also take this opportunity to verify the accuracy of my code. To do so, I will compare the results generated by my code with those obtained using the XFLR 5 software, focusing particularly on the C_p curve for a given angle of attack. Additionally, I will compare my results with thin airfoil theory, which provides a useful baseline. Indeed, for a symmetric airfoil, where the camber is zero, the lift coefficient can be expressed as:

$$C_L = 2\pi\alpha \quad (1)$$

for small angles of attack. Therefore, it is crucial to limit the angle of attack to just a few degrees to ensure the theory's validity.

To begin, let us examine the results for the symmetric rear airfoil, the NACA 0012. For this airfoil, the lift coefficient C_L is theoretically zero at a zero angle of attack. The results I obtained align with this expectation:

$$\alpha = 1^\circ \Rightarrow C_L = 0.1207 \quad (2)$$

$$\alpha = 2^\circ \Rightarrow C_L = 0.2414 \quad (3)$$

Next, we consider the cambered front airfoil, the NACA 2408. Unlike the symmetric airfoil, the cambered profile produces a non-zero lift coefficient at zero angle of attack. In this case, I obtained $C_L = 0.2465$. This result is consistent with the theoretical prediction that camber contributes to lift even in the absence of an angle of attack.

At first glance, these results suggest that my code is producing reasonable and relevant outcomes.

To confirm the accuracy of my code, I compared my results with those obtained from XFLR 5. For both airfoils and at a zero angle of attack, the following C_p plots were generated. My results are displayed on the left, while the software results are shown on the right.

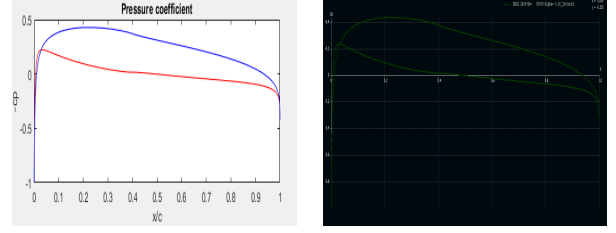


Figure 2: C_p plot for the NACA 2408.

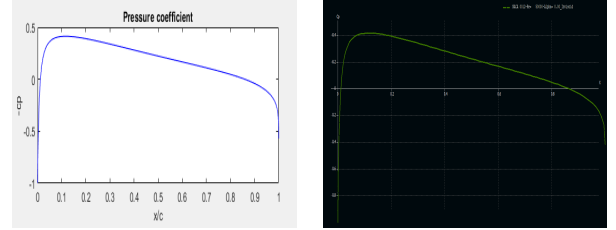


Figure 3: C_p plot for the NACA 0012.

As we can see from Figures 2 and 3, my code produces results that are consistent with those generated by the software, particularly for small angles of attack. However, it is important to note that the inviscid flow approximation used in my code has its limitations, especially at higher angles of attack.

To further investigate these limitations, I examined the divergence between my results and those accounting for viscous effects.

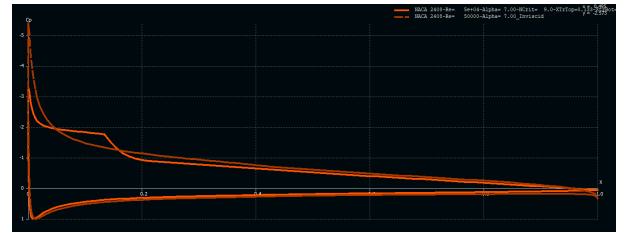


Figure 4: Divergence of C_p plots due to the inviscid approximation.

In Figure 4, the dashed line represents the C_p plot obtained using the inviscid approximation, while the solid line corresponds to the plot that includes viscous effects. For angles of attack greater than 6° , the difference between the two becomes significant. This demonstrates that the inviscid approximation is no longer valid at higher angles of attack.

Given this observation, I have decided to limit my tests to angles of attack equal to or below 6° to ensure the accuracy and reliability of my results.

4. Study for tandem configuration

4.1. No ground effect

Now that we studied both airfoil separately, we can study them on a tandem configuration. This configuration will modify the behavior of the wing, due to the presence of the tail, and vice-versa. I used the data from the assignment pdf, but these datas can be modified(see the last section)with the horizontal distance between the airfoil $x_{1,2} = 1.5m$ and the vertical distance $y_{1,2} = 0.07m$.

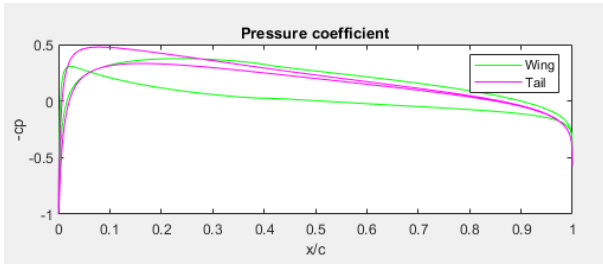


Figure 5: C_p plot for both airfoil on tandem configuration with $\alpha_w = \alpha_t = 0$

Since, I used the same angle of attack as single airfoil configuration, one can compare each results. If there is no perturbation due to the presence of the other airfoil, the C_p plot should be , for both airfoil, the same as Figure 2 and Figure 3. Or, one can see that the plot is modified and indicate that the airfoil are perturbed by each others.

4.2. Ground effect

The ground effect is a beneficial aerodynamic interaction near the ground that reduces drag and increases lift. The air between the wing and the ground is compressed, increasing the pressure under the wing. This increases lift for the same speed and angle of incidence. In addition, close to the ground, air movement under the wing is restricted, reducing vortex formation. This reduces induced drag, a major component of lift-induced drag.

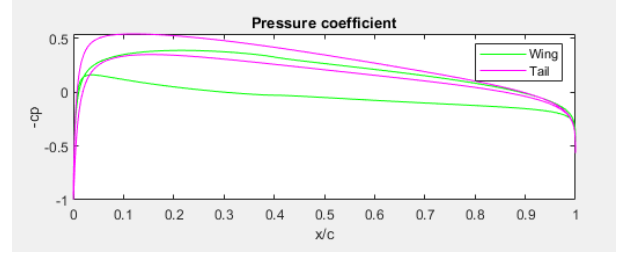


Figure 6: C_p plot for both airfoil on tandem configuration with $\alpha_w = \alpha_t = 0$ with ground effect

One can see that the C_p plot is different from Figure 4. It is due to the ground effect as explained above. The ground will compress the fluid below on the wing, modify the pressure coefficient and the lift coefficient.

5. Optimisation of the lift coefficient

The goal of aircraft design is to optimize the lift, stability, etc by changing aerodynamics values such as the angle of attack. In this section, I will fixed α_t at $\alpha_t = -0.5$ and vary α_w between 0 and 6° . I will also optimize with ground effect ($Gr_e = 1$) and without ($gr_e = 0$).

	$\alpha_w = 0$	$\alpha_w = 1$	$\alpha_w = 2$	$\alpha_w = 3$	$\alpha_w = 4$	$\alpha_w = 5$	$\alpha_w = 6$
$C_l Gr_e = 0$	0.1582	0.25	0.348	0.444	0.5413	0.6383	0.735
$C_l Gr_e = 1$	0.2316	0.378	0.517	0.648	0.772	0.889	1.008

Table 2: C_l at $\alpha_t = -0.5$ and $\alpha_w \in [0, 6]$

The best lift coefficient is for $\alpha_w = 6$. One can expected this result because of the linear dependance of C_l and α . I can now fix this value of $=\alpha_w$ and vary $\alpha_t \in [-2.5, 0.5]$

	$\alpha_t = -2.5$	$\alpha_t = -1.5$	$\alpha_t = -0.5$	$\alpha_t = 0$	$\alpha_t = 0.5$
$C_l Gr_e = 0$	0.614	0.6770	0.7355	0.7647	0.7938
$C_l Gr_e = 1$	0.9582	0.9799	1.008	1.01	1.02

Table 3: C_l at $\alpha_t = -0.5$ and $\alpha_w \in [0, 6]$

Again, the best lift coefficient is for $\alpha_t = 0.5$. As the previous case, one should expect these results because of the linear dependance of C_l for small angle of attack, this is the case here to keep the code realistic. One can now plot the pressure coefficient with the optimal angles α_t and α_w with and without ground effect.

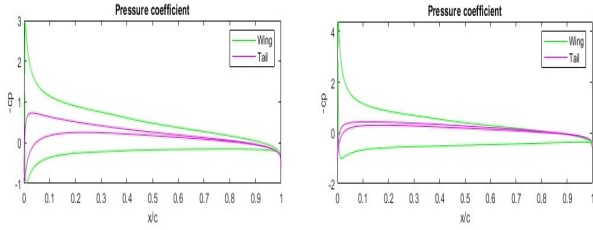


Figure 7: C_p optimal without ground effect (left) and with ground effect (right)

6. Real indoor aircraft

In this section, I will use the characteristics of a real indoor model aircraft to compute the lift coefficient and the most optimal one. For this, I choose the Cessna 172. The front airfoil profile is the NACA 2412 and for the rear airfoil profile is the NACA 0012. I found that $x_{1,2} = 0.45m$ and $y_{1,2} = 0.2m$ [2]. I fixed the angle of attack of the tail, $\alpha_t = 0.5^\circ$ and compute the lift coefficient of the front wing

	$\alpha_w = 0$	$\alpha_w = 1$	$\alpha_w = 2$	$\alpha_w = 3$	$\alpha_w = 4$	$\alpha_w = 5$	$\alpha_w = 6$
$C_l \text{ } Gr_e = 0$	0.8987	1.1517	1.4163	1.6926	1.9810	2.2819	2.5959
$C_l \text{ } Gr_e = 1$	0.8234	1.1321	1.4387	1.7435	2.0467	2.3487	2.6496

Table 4: C_l at $\alpha_t = 0.5$ and $\alpha_w \in [0, 6]$

I'm quite surprised by these results because when I add the ground effect, I get a decrease in the lift coefficient, which is contrary to previous results. Perhaps these results are due to the different distances reduced between the wing and the tail.

I can still plot the optimal C_p

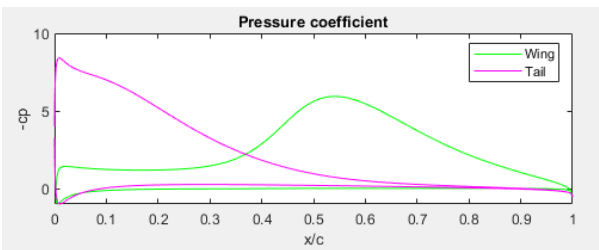


Figure 8: Optimal C_p

7. Bibliography

- [1]: Fundamental Mechanics of Fluids , I.G Currie
- [2]: Wikipedia : Cessna 172

8. Conclusions

This study investigated the aerodynamic interactions of two NACA airfoils (2408 and 0012) in a tandem configuration, focusing on maximizing the lift coefficient (C_L) of the front airfoil. Results showed that the rear airfoil significantly impacts the flow around the front airfoil, with ground effect further enhancing lift and reducing drag. Optimal angles of attack ($\alpha_t = 0.5^\circ, \alpha_w = 6^\circ$) were identified, aligning with theoretical predictions. While the inviscid model provided valuable insights, future studies should address viscous effects and explore alternative configurations for greater applicability.



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3D designing of a finite wing

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are optimized. The project evaluates the influence of the ground effect.

This report details the methodology, computational setup, results, and technical discussions. Comparisons with theoretical expectations and validation using XFLR5 are included to ensure the reliability and accuracy of the findings.

Config	CPU	RAM	Clock time
	Intel core i3-6006U	12Go	2.0 GHz

Table 1: Configuration of my computer

1. Introduction

The 3D project outlined in this report focuses on the design and optimization of a finite wing to achieve specific aerodynamic targets. The main objectives include designing a wing that generates a lift coefficient (c_L) of 0.28 ± 0.05 at a fixed angle of attack ($\alpha_w = 2.3^\circ$), while minimizing the induced drag. Subsequently, a tail section is added to the configuration to minimize the pitching moment ($|c_{m,i}|$) of the entire system. The analysis also accounts for the effects of ground proximity on aerodynamic performance.

To meet these objectives, the wing's geometric parameters, such as aspect ratio (AR), taper ratio (λ), sweep angle (Λ), and dihedral angle (d),

2. Guidelines

This project follows a structured approach to design and optimize a finite wing for specific aerodynamic targets. The primary objective is achieving a lift coefficient C_L of 0.25 ± 0.05 at a fixed angle of attack ($\alpha_w = 2.3^\circ$) while minimizing the induced drag coefficient C_d . Key geometric parameters, including aspect ratio, taper ratio, wingspan, sweep angle, dihedral angle, and chord length, are systematically analyzed and optimized. Simulations are conducted using 3D computational models, validated with XFLR5, and account for the ground effect to ensure realistic results. Practical constraints are incorpo-

rated to balance performance and manufacturability.

3. Building and Optimization of the front wing

The goal of this section is to build and optimize the wing by ensuring a lift coefficient $C_L = 0.28 \pm 0.05$ and the minimum induced drag coefficient C_d . Indeed, since my simulation is in 3D, the presence of vortices on the root and tip wing modify the lift coefficient and create the induced drag. There is an infinite combination to reduce the drag coefficient but I will, in the following part, study the effect of every characteristics but fixing the others.

Taper Ratio	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{2}{3}$	1
C_d	0.0199	0.0198	0.0195	0.0192	0.0186
C_L	0.2372	0.2367	0.2331	0.2297	0.2222

Table 2: Effect of taper ration on C_d and C_L

One can see that, the greater taper ratio is, smaller are the induced drag and lift coefficient. After $T_R > 6/10$, the lift coefficient is not in the targeted values. so I am still between the targeted values. A smaller taper ration will increase the lift coefficient but is harder to manufacture.

Dihedral angle	$d = -1$	$d = 0$	$d = 1$	$d = 2$	$d = 4$
C_d	0.0197	0.0197	0.0197	0.0197	0.0196
C_L	0.2349	0.2349	0.2349	0.2348	0.2348

Table 3: Effect of dihedral angle on C_d and C_L

One can see that the dihedral do not affect the lift and drag coefficient. The theoritical lift coefficient provided by airplane performance [1] is :

$$C_l = C_{l_\alpha}(\alpha + \Gamma \beta + i^w(y)) \quad (1)$$

If $\beta = 0$, the dihedral has no effect on the lift coefficient. One can conclude that my code gives realistic results, according to the theory.

Sweep angle	$\Lambda = 0$	$\Lambda = 2$	$\Lambda = 4$	$\Lambda = 6$	$\Lambda = 8$
C_d	0.0197	0.0197	0.0197	0.0196	0.0195
C_L	0.2357	0.2356	0.2351	0.2345	0.2336

Table 4: Effect of sweep angle on C_d and C_L

One can remark that the lift coefficient decreases when I increase the sweep angle.

Wingspan	$W_s = 10$	$W_s = 15$	$W_s = 20$	$W_s = 25$	$W_s = 30$
C_d	0.0182	0.0192	0.0197	0.02	0.0202
C_L	0.2179	0.229	0.2348	0.2384	0.2409

Table 5: Effect of wingspan on C_d and C_L

One can see that the longer is the wing, the greater is the lift coefficient. For the latter, I will take $W_s = 20$ since bigger wingspan is difficult to build and may cause structural issues.

Chord	$c = 0.5$	$c = \frac{2}{3}$	$c = 1$	$c = \frac{3}{2}$	$c = 2$
C_d	0.0204	0.0202	0.0197	0.0189	0.0182
C_L	0.2439	0.2409	0.2348	0.2261	0.2179

Table 6: Effect of chord on C_d and C_L

One can see that the shorter is the chord, the greater is C_L . For the latter, I will take the chord $c = 1$, since it is difficult to manufacture. To minimize the induced drag coefficient C_d , I choose these parameters : $T_R = 0.4, W_s = 20, c = 1, d = 2, SA = 5$.

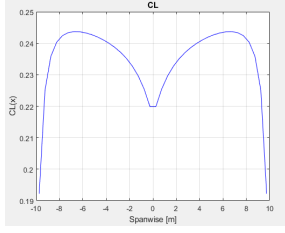


Figure 1: C_L plot from my code for the main wing only

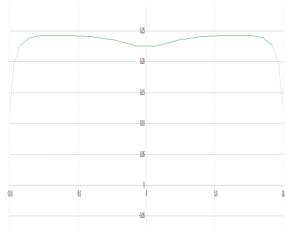


Figure 2: C_L plot from XFLR5 for the main wing only

One can see that the two different plots are a little different, especially around the fuselage, where the lift coefficient is smaller from my results.

4. Tandem configuration

Now that I have my optimized drag coefficient, I can add a tail to optimize my pitching moment, around the b_2 axis from the body reference, and keeping my lift coefficient between the targeted values. Since I put both bodies in tandem configuration, the lift coefficient of my wing will be modified, so to compare evenly the results, I have to keep my lift coefficient between the targeted values. In this section, I will also add ground effect, and I expect that my lift coefficient will increase, as my induced drag coefficient.

4.1. Effect of the characteristics of the tail on the pitching moment

Wingspan	$W_s = 4$	$W_s = 6$	$W_s = 8$	$W_s = 10$	$W_s = 12$
$ C_m $	0.0206	6×10^{-4}	0.008	0.01	0.0116

Table 7: Effect of wingspan on $|C_m|$

One can see that the pitching is not linearly related to the wingspan. A wingspan greater than 12 is not very relevant or realistic for usual air-

crafts.

Chord	$c = 0.5$	$c = \frac{2}{3}$	$c = 1$	$c = \frac{3}{2}$	$c = 2$
$ C_m $	0.23	0.013	6×10^{-4}	0.07	0.1065

Table 8: Effect of chord on $|C_m|$

Like for the wingspan, the pitching moment is not linearly related to the chord. I choose a chord equal to 1 because of the simplicity for any others computations.

Taper Ratio	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{2}{3}$	1
$ C_m $	0.037	Nan	0.02	0.015	6×10^{-4}

Table 9: Effect of taper ratio on $|C_m|$

One can remark that for $T_R = \frac{1}{4}$, the pitching moment is not defined. I wonder why my code is producing a non defined result, but I couldn't find any solution to solve my issue. A taper ratio $T_R = 0.25$ for the tail is realistic but a greater taper ration is more likely to happen.

Angle of attack	$\alpha = -2$	$\alpha = -1$	$\alpha = 0$	$\alpha = 1$	$\alpha = 2$
$ C_m $	0.83	0.4	6×10^{-4}	0.4	0.8

Table 10: Effect of angle of attack on $|C_m|$

The lift is proportional to the angle of attack for a symetric airfoil. Furthermore, the pitching moment is proportional to the lift on the horizontal tail, since the pitching moment is equal to the lift multiplied by the distance between the aerodynamic center of the tail and the aerodynamic center of the aircraft. Thus, the theory provides a pitching proportional to the angle of attack, results confirmed by my results. [1]

Sweep angle	0	2	4	6	8
$ C_m $	6×10^{-4}	3.9×10^{-4}	0.0013	0.0021	0.0027

Table 11: Effect of Sweep angle on $|C_m|$

Dihedral angle	0	2	4	6	8
$ C_m $	3.9×10^{-4}	0.0017	0.004	0.006	0.0085

Table 12: Effect of dihedral angle on $|C_m|$

One can remark that the pitching moment is increasing with the sweep angle and the dihedral angle. A small positive sweep angle and dihedral for the tail are quite common for commercial aircraft and can be justified by the value of the pitching moment

4.2. No ground effect

The tail will modify the pitching moment. Indeed, my tail will create a lever arm, and will create a pitching moment thanks to the lift force that apply on the surface of the tail.

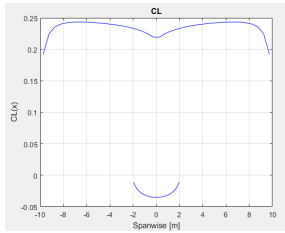


Figure 3: C_L plot from my code without ground effect

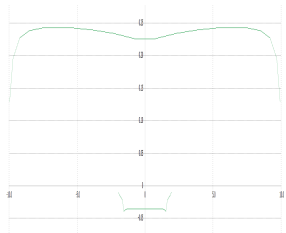


Figure 4: C_L plot from XFLR5 without ground effect

One can see that the two plot are quit similar, except for the lower part, where my code produces a simplified curve. Since the difference is not too big, I consider that my results are still realistic and match results from XFLR5. One can also see that the C_L is higher than the C_L from a wing only, as expected. As I will deal with it later, the induced drag coefficient will also be modified. Indeed, this coefficient is "related" to the lift coefficient, so if I increase my lift coefficient, I will also increase my lift coefficient. For exemple, $C_{Di} = \frac{C_L^2}{\pi A_R}$. These results should be treated with caution because of some assumptions made but it gives a good insight of the relation between lift and induced drag.

4.3. Ground effect

The ground effect is a beneficial aerodynamic interaction near the ground that reduces drag and increases lift. The air between the wing and the ground is compressed, increasing the pressure under the wing. This increases lift for the same speed and angle of incidence. In addition, close to the ground, air movement under the wing is restricted, reducing vortex formation. This reduces induced drag, a major component of lift-induced drag.

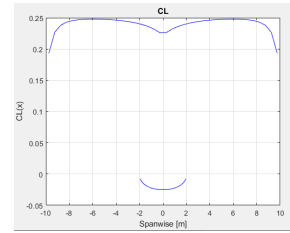


Figure 5: C_L plot from my code with ground effect

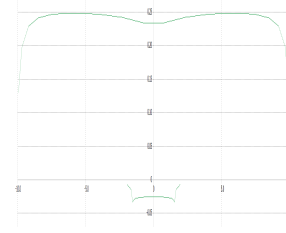


Figure 6: C_L plot from XFLR5 with ground effect

One can see that the lift coefficient seems higher than the C_L from no ground effect but it is nearly insignificant.

5. Conclusions

The thesis project explores the design and optimization of a finite wing to meet specific aerodynamic objectives. These include achieving a target lift coefficient of 0.28 ± 0.05 at a fixed angle of attack, while minimizing induced drag. The study involves a systematic investigation of key geometric parameters, such as aspect ratio, taper ratio, sweep angle, and dihedral angle, to determine their influence on performance. Computational simulations, validated against theoretical predictions and results from XFLR5 software, form the backbone of the analysis. One significant finding is the role of ground ef-

fect in improving aerodynamic efficiency. When operating near the ground, the wing experiences increased lift and reduced induced drag, attributed to compressed airflow beneath the wing and limited vortex formation. These insights highlight the importance of accounting for operational conditions in aerodynamic design.

In addition to the main wing design, a tail section was introduced to optimize the pitching moment without compromising the target lift coefficient. The tandem configuration posed challenges, such as maintaining consistent aerodynamic properties across the system, but provided valuable results for practical aircraft design considerations.

References

[1] Carlo E.D Riboldi : Flight dynamics Modeling, characterization and performance